The Mathematical Thinking Process

The Mathematical Association of Western Australia Inc.
MAΣWA

Based on the Mathematical Thinking Process Model outlined in SCSA syllabus

1 Interpret the task and gather the key information

- What is the purpose of the task?
- What information has been provided with this task?
- What information will I need to pursue?
- Upon completion of this task: what decisions can be made and what conclusions can be stated?

2 Identify the mathematics which could help to complete the task

- What mathematical skills and concepts could I use to work on this task?

3 Analyse information and data from a variety of sources

- Upon analysis of information, do I have enough, or do I need to research for further information?
- What are the best sources to obtain the required information?

4 Apply exisiting mathematical knowledge and strategies to obtain a solution

- How do I apply the identified mathematical processes to complete this task?
- How do I solve the task using the concepts and skills identified above?
- Have I become aware of other processes that will assist in the solution to the task?

5 Verify the reasonableness of the solution

With the original task in mind:

- Is the solution valid?
- Does this solution solve the task?
- Is the solution both, efficient and optimal?

6 Communicate findings in a systematic and concise manner

- Have I explained the solution to the task in an efficient way using appropriate language with diagrams if necessary, referring to the original task, directing the conclusion/s to the target audience and referenced my data source/s?